

ative to the default (non-persona) setting: ingroup sentences were $6.74\times$ more likely to express ingroup solidarity, while outgroup sentences were $1.50\times$ more likely to exhibit outgroup hostility.

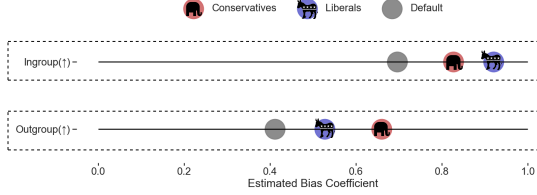


Figure 5: Estimated significant coefficients of Ingroup bias (higher the coefficient higher ingroup solidarity) and outgroup bias (higher the coefficient higher the outgroup hostility) per LLMs. We estimate these changes with respective regression coefficients for three persona (conservatives, Liberal and default) where ingroup bias score is the ‘us versus them’ score of “we are” sentences and outgroup bias is “they are” sentences. All estimated bias coefficients are significant (p-values below 0.001). The differences in the numbers of sentences per LLMs do not affect these results, since the estimates are computed per sentences.

Prior work often relies on political orientation tests (e.g., the Political Compass) to diagnose model ideology, but such instruments can miss the finer dynamics of bias in LLM-generated text (Rozado, 2024; Liu et al., 2021). Building on our finding that persona induction amplifies both ingroup solidarity and outgroup hostility, we directly modeled sentence-level bias by regressing bias scores separately for ingroup (“We are”) and outgroup (“They are”) bias across three personas (Conservative, Liberal, and Default). As shown in Fig. 5, persona indicators are strong, statistically significant predictors of bias. For ingroup solidarity, Liberal-persona sentences are 11.26% higher than Conservative ($p < 0.001$) and 32.25% higher than Default. For outgroup hostility, both Conservative and Liberal personas show significant increases, with the Conservative persona 25.0% higher than Liberal and 60.2% higher than Default. Taken together, these results indicate that persona conditioning systematically shifts ‘us versus them’ framing—elevating ingroup positivity for the Liberal persona while intensifying outgroup negativity for the Conservative persona. This persona-sensitive, sentence-level analysis complements orientation-level assessments in the literature (Liu et al., 2021; Rozado, 2024; Bang et al., 2024), which often find LLM outputs to lean so-

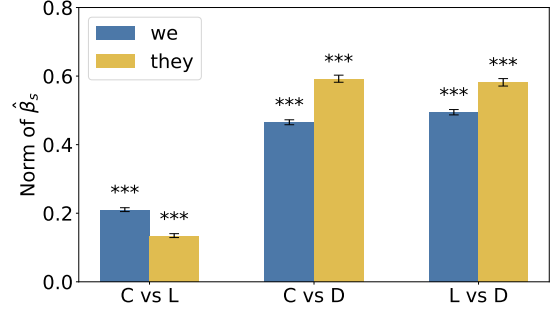


Figure 6: Differences in word meaning across pairwise persona comparisons: C vs L (Conservative vs Liberal), C vs D (Conservative vs Default), and L vs D (Liberal vs Default). Reported values correspond to the norm of β coefficients with bootstrap confidence intervals. *** indicates statistical significance at the 0.01 level.

called ‘left-of-center’, by revealing how identity cues modulate the direction and magnitude of social identity bias.

At the persona level, to quantify model-specific us versus them bias differences, we regressed sentence-level bias scores for ingroup and outgroup sentences across LLMs (see Appendix A.7). Ingroup effects are consistently positive yet tightly clustered: relative to DeepSeek-3.1, GPT-4.1 shows a +1.9% increase, Grok-3.0 +3.2%, LLaMA-3.1 +4.6%, and Gemma-2.0 exhibits the largest uplift at +14.9%, making it the most ingroup-affirming model, while DeepSeek-3.1 remains the least. In contrast, outgroup hostility shows greater divergence across models. Compared to DeepSeek-3.1, GPT-4.1 registers a +163% increase, Gemma-2.0 +176%, Grok-3.0 +199%, and LLaMA-3.1 a striking +283%, positioning LLaMA-3.1 as the most outgroup-hostile model and DeepSeek-3.1 as the least. The resulting coefficients are presented in Table 14.

To examine how persona conditioning alters the internal representation space of LLMs, we applied t-SNE to project sentence embeddings into two dimensions and compared the resulting distributions across LLMs. As shown in Appendix A.5, Figure 20, embeddings of sentences generated with political personas versus the default (no-persona) setting form distinct clusters, indicating that persona information systematically reshapes the models’ representational geometry. All models exhibit clear separation between persona conditions, suggesting that persona signals are consistently encoded within their embedding spaces.

Table 2: Top 10 Nearest Neighbors (NN) and Nearest Contexts (NC) for the Target Terms “We” and “They” in the Embedding Space of Each Persona LLM-Generated Corpus. The table shows the lexical and contextual similarities of the target terms across conservative, liberal, and default personas.

Persona	NN _{We}	NN _{They}
Conservatives	upholding, staunch, uphold, conservative, conservatives, advocate, supporter, liberal, ardent, liberals	hypocritical, advocating, espousing, ideology, undermining, respecting, intolerant, championing, betraying, liberalism
Liberals	advocate, equality, advocacy, fostering, advocating, championing, advocates, upholding, safeguarding, promoting	advocating, championing, hypocritical, appeasing, accuse, espousing, marginalizing, espoused, intolerant, ideologues
Default	imagining, dreaming, rediscovering, longing, dreamers, souls, imagined, fascinated, immortal, dreams	lovely, gorgeous, beautiful, dancing, tossing, multicolored, shiny, poking, dreaming, bobbing

By applying an embedding regression technique, we assess how lexical items (e.g., “we,” “they”) are semantically differentiated in partisan discourse. This approach moves beyond frequency-based analyses by quantifying contextual divergence in word usage and linking it directly to political covariates. To measure how the meanings of the focal pronouns “we” and “they” vary across personas, we estimated pairwise semantic shifts. As shown in Figure 6, both “we” and “they” exhibit substantial divergence from the Default condition for conservative and liberal (0.46 & 0.50) for “we” and (0.58 & 0.59) for “they.” In contrast, the semantic difference between C vs D is smaller, at 0.21 for “we” and 0.13 for “they.” These findings align with our bias scoring analysis, showing that persona conditioning meaningfully alters both the embedding space and bias magnitude.

Table 2 shows the lexical neighborhoods for “we” and “they” in embedding space of our corpus. Conservative usage emphasizes solidarity and protection (upholding, staunch, steadfast, conservative) whereas Liberal highlights civic action and rights (advocate, equality, safeguarding, promoting, rights). In the Default (no-persona) condition, we detaches from partisan identity and clusters with affective, imaginative terms (imagining, dreaming, longing, souls, immortal). Similarly For the term

“they”, Conservative contexts depict a threatening outgroup (hypocritical, espousing, undermining, ideology, intolerant), Liberal contexts frame they in social-change discourse(advocating, championing, marginalizing). In the default (no-persona) persona settings “they” is non-political associates with descriptive, aesthetic vocabulary (lovely, gorgeous, colorful, multicolored). Together, these patterns corroborate that “we” functions as an ingroup marker (loyalty vs. rights depending on persona), whereas “they” functions as an outgroup marker (threat vs. reform). We corroborate this by examining the top nearest contexts—specifically, the individual contexts of the terms “We” and “They,” embedded using ALC, that are closest to each persona’s ALC embedding of the respective term (see Appendix A.8, Table 15).

Also, Figure 7 shows the difference in average sentiment (Φ_{avg}) between liberal and conservative corpora. Liberal persona sentences exhibit higher Φ_{avg} due to more frequent positive and fewer negative words. In contrast, conservative outputs contain more negative terms (− ↑) such as fighting, poverty, violence, harm, crisis, and unfair, and fewer positive ones (+ ↓) like freedom, liberty, faith, and proud, lowering Φ_{avg} relative to liberals. Some words show the opposite pattern—positive terms like health, energy, and profit (+ ↑) appear more in conservative text, while negative ones such as fear, guilt, and shame (− ↓) occur more in liberal outputs. Overall, the 0.25 decrease in Φ_{avg} for the conservative corpus stems mainly from a higher prevalence of negative and fewer positive words.

We also compared the lexical polarization between liberal and conservative persona using allotaxonomic rank-rank histogram between liberal and conservative corpora shows in Figure 8 presents. Clear lexical and thematic divergences emerge. Terms such as ‘rights’ (15th), ‘social’ (17th), and ‘justice’ (23rd) rank highly in the liberal corpus but are substantially down-ranked in the conservative corpus, reflecting a stronger emphasis on equity and justice-related themes. High-frequency stopwords (‘we’, ‘our’, ‘and’, ‘the’, ‘for’, ‘to’) dominate both corpora and contribute substantially to divergence. Liberal-exclusive words such as ‘equity’, ‘belief’, ‘tireless’, ‘affordable’, and ‘reproductive’ reinforce a focus on social justice and healthcare, though their lower frequency limits their contribution to the divergence measure. By contrast, the conservative corpus is characterized by terms linked to governance, tradition, and

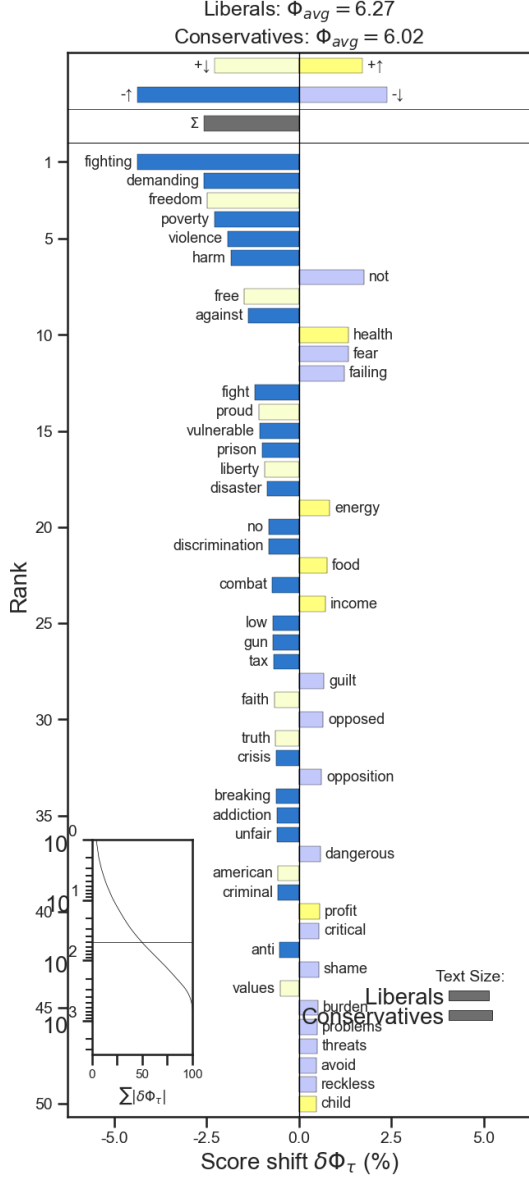


Figure 7: Word shift graph comparing conservative and liberal persona corpora. Words are ranked by their percentage contribution to the change in average happiness (Φ_{avg}). The liberal corpus serves as the reference text (T_{ref}), while the conservative corpus is used as the comparison text (T_{comp}).

national identity. Words such as ‘government’, ‘defense’, ‘freedom’, ‘brave’, ‘national’, and ‘enterprise’ appear prominently, underscoring themes of authority, security, and sovereignty largely absent from the liberal corpus. Overall, the divergence highlights, liberal outputs foreground social justice, healthcare, and anti-racism discourse, while conservative outputs emphasize governance, defense, and national identity.

4.3 Case Study: Performance of the Open Language Model (OLMoE)

Table 3: Ingroup solidarity ($\mu_{ingroup}$) and outgroup hostility ($\mu_{outgroup}$) across persona for open language model (OLMoE).

Persona	In-group	Out-group
Default	20.03	12.66
Conservative	41.54	22.05
Liberal	20.03	20.05

To further assess us–them bias in open-source systems, we evaluated OLMoE (OLMoE-1B-7B) (Muennighoff et al., 2024), a fully transparent Mixture-of-Experts (MoE) language model, using the same prompts described in Appendix A.2. Compared to the average of other large language models (LLMs), OLMoE exhibited markedly lower bias across all conditions. In the default (non-persona) setting, its in-group bias was 70.6% lower and out-group bias 78.2% lower than the mean of other models. When conditioned on political personas, OLMoE remained notably balanced, with bias scores 45–76% lower than the LLM average shows in Table 3, though slight liberal bias persists.

We attribute this reduction to OLMoE’s non-social-media pretraining corpus, composed exclusively of factual and domain-specific sources (DCLM-Baseline web corpus, StarCoder code data, peS2o and arXiv papers, OpenWebMath, Algebraic Stack, and English Wikipedia/Wikibooks) over 4 trillion tokens (Muennighoff et al., 2024). The absence of conversational or ideologically charged data likely contributes to OLMoE’s exceptional neutrality, highlighting the importance of transparent, domain-curated datasets for mitigating (or at least, predicting) social bias in open LLMs.